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Identical transformation of fractional-rational expressions

Everything from §§2–5 in one question: the long chains where order of operations decides whether the work takes three lines or thirteen.

LESSON 20–22 | 3 × 40 MIN | ALGEBRA 8, §6 | STAGE 9 · 2.3, 2.6

LESSON CLOCK

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Simplify a fractional-rational expression that mixes all four operations.
- Deal with a compound (“stacked”) fraction.
- Choose an order of work that keeps the expressions small.
- Prove a given identity by transforming one side into the other.

TEXTBOOK

National	§6, pp. 30–33
Cambridge	Learner’s Book pp. 30–39, 45–53
Workbook	Workbook 2.6

01 Explanation

What “identical transformation” means

DEFINITION

Two expressions are **identically equal** if they take the same value for every permissible value of the variables. Replacing an expression by an identically equal one is an **identical transformation**.

Cancelling, building up to a common denominator, adding, multiplying — each of these is an identical transformation. That is exactly why the permissible values must be stated: outside them the two expressions are *not* the same, because one of them does not exist.

Order of operations

The order is the ordinary one — brackets, then multiplication and division, then addition and subtraction — but with fractions it pays to look ahead:

1. **Tidy every bracket first** into a single fraction.

2. Do the multiplications and divisions, cancelling as you go.
3. Do the additions and subtractions last, over the LCD.
4. Cancel the final answer. Check the permissible values against the *original*.

> THE LONG WAY ROUND
>
> Multiplying everything out at the start turns a two-line problem into a page of algebra. Almost every expression in this section is designed to collapse — if your working is growing, you have taken the wrong route.

Compound fractions

A fraction whose numerator or denominator is itself a fraction is a **compound fraction**. The main fraction bar means “divide”:

$$\frac{\frac{1}{x} - \frac{1}{y}}{\frac{1}{x} + \frac{1}{y}} = \left(\frac{1}{x} - \frac{1}{y}\right) : \left(\frac{1}{x} + \frac{1}{y}\right)$$

Combine the top into one fraction, combine the bottom into one fraction, then divide:

$$\frac{y - x}{xy} : \frac{y + x}{xy} = \frac{y - x}{xy} \cdot \frac{xy}{y + x} = \frac{y - x}{y + x}$$

Proving an identity

To prove $A = B$, transform the **more complicated** side until it becomes the simpler one. Never move terms across the equals sign as if it were an equation — you are not solving, you are simplifying.

02 Worked examples

EXAMPLE 1 ***Simplify*** $\dfrac{\mathbf{a+b}}{\mathbf{a-b}} - \dfrac{\mathbf{a-b}}{\mathbf{a+b}}$

1. $LCD = (a - b)(a + b) = a^2 - b^2$

   Different factors, so multiply them.

2. $$\frac{(a+b)^2 - (a-b)^2}{a^2 - b^2}$$

   Build both up. Keep the bracket on the subtracted square.

3. $(a + b)^2 - (a - b)^2 = (a^2 + 2ab + b^2) - (a^2 - 2ab + b^2) = 4ab$

   The squares cancel; only the middle terms survive.

4. $$\frac{4ab}{a^2 - b^2}$$

   Nothing left to cancel.

ANSWER

$$\frac{4ab}{a^2 - b^2}, \quad a \neq \pm b$$

EXAMPLE 2 **Simplify** $\dfrac{1-\dfrac{1}{x}}{1-\dfrac{1}{x^2}}$

1. $1-\dfrac{1}{x}=\dfrac{x-1}{x}$

   Top, as one fraction.

2. $1-\dfrac{1}{x^2}=\dfrac{x^2-1}{x^2}$

   Bottom, as one fraction.

3. $\dfrac{x-1}{x}:\dfrac{x^2-1}{x^2}=\dfrac{x-1}{x}\cdot\dfrac{x^2}{(x-1)(x+1)}$

   Divide, and factorise x^2-1.

4. *cancel $(x-1)$ and one x*

5. $\dfrac{x}{x+1}$

ANSWER

$\dfrac{x}{x+1}$, $x \neq 0, x \neq \pm 1$

EXAMPLE 3 ***Prove that*** $\left(x + \frac{1}{x}\right)^2 - \left(x - \frac{1}{x}\right)^2 = 4$

1. Use $A^2 - B^2 = (A - B)(A + B)$ with $A = x + \frac{1}{x}$ and $B = x - \frac{1}{x}$.

   Far quicker than expanding both squares.

2. $A - B = \frac{2}{x}$

   The x terms cancel.

3. $A + B = 2x$

   The 1/x terms cancel.

4. $\frac{2}{x} \cdot 2x = 4$

   The x cancels — a constant, for every permissible x.

ANSWER

$4, \; x \neq 0$

03 Interactive model

These four are the pattern the control work will use. Walk the class through the choice of route, not just the algebra.

Long chains, one step at a time

$$\left(\frac{1}{x} - \frac{1}{y}\right) : \left(\frac{1}{x} + \frac{1}{y}\right)$$

1. $$\frac{1}{x} - \frac{1}{y} = \frac{y - x}{xy}$$

Combine the first bracket.

2. $$\frac{1}{x} + \frac{1}{y} = \frac{y + x}{xy}$$

Combine the second.

3. $$\frac{y - x}{xy} \cdot \frac{xy}{y + x}$$

Division becomes multiplication by the reciprocal.

4. $$\frac{y - x}{y + x}$$

The whole of xy cancels.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Rational expression	Ratsional ifoda	Рациональное выражение
Fractional-rational expression	Kasr-ratsional ifoda	Дробно-рациональное выражение
Compound fraction	Murakkab kasr	Многоэтажная дробь
Identity	Ayniyat	Тождество
Identically equal	Ayniy teng	Тождественно равны
Prove	Isbotlash	Доказать
Order of operations	Amallar tartibi	Порядок действий
Transform	Almashtirish	Преобразовать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\frac{a}{a-b} + \frac{b}{b-a}$ equals:

- 0
- 1
- $\frac{a+b}{a-b}$
- -1

To simplify a compound fraction you should:

- cancel the small fractions first
- combine the top and the bottom, then divide
- multiply everything out
- add the numerators

$(x + \frac{1}{x})^2 - (x - \frac{1}{x})^2$ equals:

- 0
- 2
- 4
- $4x^2$

Proving an identity means:

- solving for x
- transforming one side into the other
- moving terms across the equals sign
- substituting one value of x

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\left(\frac{1}{x} + \frac{1}{y}\right) \cdot xy$

ANSWER $y + x$

2. $1 + \frac{1}{x}$

ANSWER $\frac{x + 1}{x}$

3. $\frac{a}{b} + 1$

ANSWER $\frac{a + b}{b}$

4. $\frac{x}{2} + \frac{x}{3}$

ANSWER $\frac{5x}{6}$

5. $1 - \frac{1}{2x}$

ANSWER $\frac{2x - 1}{2x}$

6. $\frac{x + 1}{2} \cdot \frac{4}{x + 1}$

ANSWER 2

7. $\frac{2}{x} : \frac{4}{x^2}$

ANSWER $\frac{x}{2}$

Medium · the standard question

7 PROBLEMS

1. $\left(\frac{1}{x} - \frac{1}{y}\right) : \left(\frac{1}{x} + \frac{1}{y}\right)$

ANSWER $\frac{y - x}{y + x}$

2. $\left(1 + \frac{1}{x}\right) \cdot \frac{x}{x + 1}$

ANSWER 1

3. $x - \frac{1}{x}$

ANSWER $\frac{x^2 - 1}{x}$

4. $\frac{a + b}{a - b} - \frac{a - b}{a + b}$

ANSWER $\frac{4ab}{a^2 - b^2}$

5. $\left(\frac{1}{x - 1} + \frac{1}{x + 1}\right) \cdot \frac{x^2 - 1}{2}$

ANSWER x

6. $\frac{x^2 - 1}{x} : (x + 1)$

ANSWER $\frac{x - 1}{x}$

7. $\left(\frac{1}{a} + \frac{1}{b}\right) : \frac{1}{ab}$

ANSWER $a + b$

Hard · stretch and reason

7 PROBLEMS

1. $\left(\frac{1}{x-1} - \frac{1}{x+1}\right) : \frac{2}{x^2-1}$

ANSWER $1, x \neq \pm 1$

2. $\left(x + \frac{1}{x}\right)^2 - \left(x - \frac{1}{x}\right)^2$

ANSWER 4

3. $\frac{a}{a-b} + \frac{b}{b-a}$

ANSWER $1, a \neq b$

4. $\dfrac{1 - \frac{1}{x}}{1 - \frac{1}{x^2}}$

ANSWER $\frac{x}{x+1}$

5. $\left(\frac{1}{x} + \frac{1}{y}\right) \cdot \frac{xy}{x+y}$

ANSWER 1

6. *Prove that* $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$

ANSWER LCD $n(n+1)$: the numerator is $(n+1) - n = 1$.

7. $\frac{1}{x^2-x} + \frac{1}{x^2-1}$

ANSWER $\frac{2x+1}{x(x^2-1)}$

07 Homework

Homework — 5 problems

Algebra 8, §6, pp. 30–33. Choose your route before you start writing.

1. $\left(\frac{1}{x} - \frac{1}{x+1}\right) \cdot x(x+1)$
2. $\frac{x+y}{x-y} + \frac{x-y}{x+y}$
3. $\frac{1 + \frac{1}{x}}{1 - \frac{1}{x}}$
4. $\left(\frac{a}{b} - \frac{b}{a}\right) : \frac{a^2 - b^2}{ab}$
5. *Prove that* $\frac{1}{x-1} - \frac{1}{x+1} = \frac{2}{x^2 - 1}$